

Dense critical graphs from saturated graphs

Qiyuan Gu
University of Chicago
phoenix1203@uchicago.edu

Abstract

Let $f_k(n)$ denote the maximum number of edges in a k -critical graph on n vertices. For every fixed integer $k \geq 8$, we construct a sequence of k -critical graphs G_s with $|V(G_s)| \rightarrow \infty$ and

$$\frac{e(G_s)}{|V(G_s)|^2} \rightarrow \begin{cases} \frac{k-4}{2(k-2)}, & k \text{ even}, \\ \frac{k-5}{2(k-3)}, & k \text{ odd}. \end{cases}$$

For every $k \geq 8$ except $k = 9$, this improves the best known lower bounds, due to Toft (1970). For k divisible by three and at least twelve, it disproves the asymptotic formula proposed in Erdős Problem 917, previously known to fail when $3 \nmid k$. The smallest new case for the formula is $k = 12$, with density $2/5 > 3/8$. The construction combines a coloring module of Pegden with saturated graphs of sublinear maximum degree. The complete proof is formalized and verified in Lean 4.

1 Introduction

All graphs are finite and simple. A graph is k -critical if it has chromatic number k and every proper subgraph is $(k-1)$ -colorable; for graphs without isolated vertices this is equivalent to requiring only that deleting any edge lowers the chromatic number. Write $e(G) = |E(G)|$, and let $f_k(n)$ be the maximum number of edges of a k -critical graph on n vertices (in the edge-deletion sense). Erdős [3, p. 28] conjectured that, for fixed $k \geq 6$,

$$f_k(n) \sim \frac{1}{2} \left(1 - \frac{1}{\lfloor k/3 \rfloor} \right) n^2. \quad (1.1)$$

This is the general asymptotic question in Erdős Problem 917 [4]. Dirac [2, p. 90] gave the $k = 6$ construction obtained by joining two odd cycles of equal order, and Erdős [3, p. 28, eq. (3)] observed that it generalizes when $3 \mid k$, giving the coefficient in (1.1). When $3 \nmid k$, Toft's constructions [8] give larger coefficients. For background on dense critical graphs, see Jensen and Toft [6, Section 5.1, pp. 97–98] and Jensen [5]. Luo, Ma, and Yang [7] noted that no constructions improving Toft's asymptotic constants had been found since 1970; their Theorem 1.1 gives upper bounds on $f_k(n)$. We give the following construction.

Theorem 1. *For every integer $k \geq 8$, there is a sequence of k -critical graphs G_s such that $|V(G_s)| \rightarrow \infty$ and*

$$\frac{e(G_s)}{|V(G_s)|^2} \rightarrow \begin{cases} \frac{k-4}{2(k-2)}, & k \text{ even}, \\ \frac{k-5}{2(k-3)}, & k \text{ odd}. \end{cases}$$

Write c_k for the density in Theorem 1. Toft's coefficients [8], in the notation of [7, Section 1], are

$$b_k = \frac{1}{2} - \frac{3}{2k - \delta_k}, \quad \delta_k = \begin{cases} 0, & k \equiv 0 \pmod{3}, \\ 8/7, & k \equiv 1 \pmod{3}, \\ 44/23, & k \equiv 2 \pmod{3}. \end{cases}$$

Corollary 2. *For every $k \geq 8$, $k \neq 9$, the density c_k exceeds Toft's coefficient b_k . Moreover, for $3 \mid k$ and $k \geq 12$, (1.1) fails. For $k = 9$, all three coefficients equal $1/3$.*

Proof. Theorem 1 and $f_k(|V(G_s)|) \geq e(G_s)$ give $\limsup_{n \rightarrow \infty} f_k(n)/n^2 \geq c_k$. For even $k \geq 8$, the inequality $c_k > b_k$ is equivalent to $k > 6 - \delta_k$; for odd $k \geq 11$, it is equivalent to $k > 9 - \delta_k$. Both inequalities hold. When $3 \mid k$, Toft's coefficient is $b_k = \frac{1}{2}(1 - 1/\lfloor k/3 \rfloor)$. Formula (1.1) would force the normalized limit, and hence the limsup, to equal b_k , contradicting the lower bound for $k \geq 12$. At $k = 9$, all three coefficients equal $1/3$. $\square$

For even $k = 2t + 2$, we combine t copies of Pegden's module [9], joining their active sets by an almost complete t -partite graph. The omitted edges are prescribed by a K_t -saturated graph H of small maximum degree [1]. In a $(2t + 1)$ -coloring, each active set needs three colors and the degree bound forces two to be private to that set. The remaining common color would give a K_t in H , a contradiction. Saturation supplies the colorings after an edge between modules is deleted. Odd k follow by joining with K_1 .

2 A coloring module

Let $r \geq 3$, let S be r -critical, and let T be $(r - 1)$ -critical. The module $U(S, T)$ consists of disjoint copies of S and T , together with an independent set

$$A = V(S) \times V(T).$$

Each $(x, y) \in A$ is adjacent to $x \in V(S)$ and $y \in V(T)$. There are no edges between S and T , and no other edges. We call A the active set and call $\{x\} \times V(T)$ the row indexed by x .

The following is the $U(r, r - 1)$ case of Pegden's module lemma [9, Lemma 2.5], stated there for triangle-free factors; the lemma holds for general critical factors S and T of the indicated chromatic numbers.

Lemma 3. *Fix a palette of r colors.*

1. *In every proper coloring of $U(S, T)$ from this palette, the active set uses at least three colors.*
2. *Given $a_0 = (x_0, y_0) \in A$ and distinct colors α, β, γ , there is a proper coloring in which the active set uses only α, β, γ , and γ occurs on the active set only at a_0 .*
3. *Given an edge e of $U(S, T)$ and distinct colors α, β , there is a proper coloring of $U(S, T) - e$ in which the active set uses only α, β .*

Proof. For (1), suppose the active colors are contained in $\{\alpha, \beta\}$. Since S uses all r colors, it contains a vertex colored α and a vertex colored β . Every active vertex in the first row must have color β , so no vertex of T can have color β . The second row similarly excludes α from T . This leaves only $r - 2$ colors for T , a contradiction.

For (2), color S so that α occurs only at x_0 . Color T with the $r - 1$ colors other than α , so that β occurs only at y_0 . These colorings exist by criticality: delete the specified vertex, color the

remaining graph with one fewer color, and give the deleted vertex its own color. Extend these colorings by

$$\text{col}(x, y) = \begin{cases} \alpha, & x \neq x_0, \\ \beta, & x = x_0, y \neq y_0, \\ \gamma, & (x, y) = (x_0, y_0). \end{cases} \quad (2.1)$$

Each active vertex differs in color from both its neighbors.

For (3), there are four possibilities for the deleted edge. If $e \in E(S)$, color both $S - e$ and T with the $r - 1$ colors other than α , and color all of A with α . If $e \in E(T)$, color $T - e$ with the $r - 2$ colors other than α, β . Choose $x_0 \in V(S)$ and color S with α only at x_0 . Color the row indexed by x_0 with β and all other rows with α .

Suppose instead that e joins an active vertex (x_0, y_0) to one of its two structural neighbors. Use the colorings of S and T from the proof of (2). If the deleted edge joins (x_0, y_0) to x_0 , use (2.1) with the color of (x_0, y_0) changed to α . If it joins (x_0, y_0) to y_0 , change that color to β . In either case the only edge that would become monochromatic is the deleted edge. $\square$

3 From saturated graphs to critical graphs

A graph is K_t -saturated if it contains no K_t , but adding any missing edge creates a K_t . Thus the common neighborhood of every pair of distinct nonadjacent vertices contains a K_{t-2} .

Proposition 4. *Let $t \geq 3$, and let H be a K_t -saturated graph on $v \geq t$ vertices with maximum degree $d < v - 1$. Let $h \geq 2t + 1$ be odd and suppose*

$$v > 2t(t - 1)hd. \quad (3.1)$$

Put $a = 2hv$. There is a $(2t + 2)$ -critical graph G with

$$|V(G)| = t(a + h + 2v) \quad (3.2)$$

and

$$e(G) \geq \binom{t}{2} a^2 \left(1 - \frac{d}{v}\right). \quad (3.3)$$

Proof. Construction. Set $\ell = 2v$ and take joins of cliques and odd cycles, as in [2, p. 90]:

$$S = K_{2t-2} \vee C_{h-2t+2}, \quad T = K_{2t-3} \vee C_{\ell-2t+3}.$$

Both cycles are odd and have length at least three. Since the join of a p -critical and a q -critical graph is $(p + q)$ -critical, S is $(2t + 1)$ -critical and T is $2t$ -critical. (Deleting a vertex or an edge within one factor saves a color there; if a joining edge xy is deleted, give x and y singleton colors in their respective factors and identify those colors.)

Take t disjoint copies $U_1, \dots, U_t$ of $U(S, T)$, with active sets $A_1, \dots, A_t$, each of size $a = h\ell$. In each copy, identify $V(T)$ with $V(H) \times \{0, 1\}$ by any bijection. An active vertex then has the form $(x, (z, \varepsilon))$; give it label z . Let π denote this labeling map. Each label occurs exactly $b = 2h$ times in each active set.

Define a t -partite graph Z on $A_1 \cup \dots \cup A_t$ by putting, for $u \in A_i$ and $w \in A_j$ with $i \neq j$,

$$uw \in E(Z) \iff \pi(u)\pi(w) \in E(H).$$

The graph G retains all the edges of the modules and, between different active sets, takes the complement of Z . There are no other edges between modules. In particular, active vertices with equal labels in different parts are adjacent in G .

We record three properties of Z . First, Z is K_t -free: the labeling map sends every clique injectively to a clique of H . Moreover, adding any missing edge between different parts of Z creates a K_t with one vertex in each part. To see this, let u, w be the endpoints of such a missing edge. If their labels z, z' differ, saturation of H gives a K_{t-2} in $N_H(z) \cap N_H(z')$. Place its labels in the other $t-2$ parts. Together with u, w , they form the required clique in $Z + uw$. If the labels coincide, say both equal z , choose a vertex $z' \neq z$ not adjacent to z ; this is possible because $d < v-1$. The same common-neighborhood clique is contained in $N_H(z)$ and gives the required clique.

Second, Z contains a K_{t-1} across any prescribed $t-1$ parts. Indeed, H has a missing edge; one endpoint together with a K_{t-2} in the common neighborhood gives a K_{t-1} in H . Its labels can be placed in any $t-1$ parts of Z .

Third, $D := \Delta(Z) = (t-1)bd = 2h(t-1)d$, so (3.1) gives $\ell > 2tD$.

The chromatic lower bound. Suppose that G has a $(2t+1)$ -coloring. Fix a part A_i and any color α . Since its copy of S uses all $2t+1$ colors, some vertex x of S has color α . The ℓ active vertices in the row indexed by x avoid α , so the pigeonhole principle and $\ell > 2tD$ give a color $\beta \neq \alpha$ occurring more than D times in that row. Applying this observation twice, we may choose two distinct colors α_i, β_i , each used more than D times in A_i .

A color used more than D times in A_i cannot occur in any other active set. A vertex of that color in another part would have to be adjacent in Z to all those more than D vertices, contrary to $\Delta(Z) = D$. Thus the $2t$ chosen colors $\alpha_1, \beta_1, \dots, \alpha_t, \beta_t$ are distinct, leaving a single color γ . The active set A_i can use only $\alpha_i, \beta_i, \gamma$, and Lemma 3(1) forces γ to occur in every part. Choosing a vertex of this color in each part gives a K_t in Z , a contradiction. Thus $\chi(G) \geq 2t+2$.

Colorings after edge deletion. Use the $(2t+1)$ -color palette

$$\alpha_1, \beta_1, \dots, \alpha_t, \beta_t, \gamma.$$

The pair α_i, β_i will be used only in A_i among the active sets. Structural vertices may use the full palette, since they have no neighbors in other modules.

Let $e = uw$ join two different active sets in G . By the first property of Z , there is a K_t in $Z + uw$ containing uw , with vertices $a_i \in A_i$. Apply Lemma 3(2) in each module, using active colors $\alpha_i, \beta_i, \gamma$ and assigning γ only to a_i in A_i . The only color shared by different active sets is γ . Its t vertices are pairwise nonadjacent in $G - e$, because they form a clique in $Z + uw$. We have obtained a $(2t+1)$ -coloring of $G - e$.

Now let e lie within U_i . Apply Lemma 3(3) to color $U_i - e$, using only α_i, β_i on A_i . Choose a K_{t-1} in Z across the other $t-1$ parts. In each of those modules, apply Lemma 3(2) with the selected vertex as the only active vertex of color γ . Once again this is a proper $(2t+1)$ -coloring of $G - e$.

These cases cover every edge of G . Recoloring one endpoint of a deleted edge with a new color gives $\chi(G) \leq 2t+2$. Since G has no isolated vertices, every proper subgraph of G lies in some $G - e$. Hence G is $(2t+2)$ -critical.

Counting vertices and edges. Each module has $a + h + 2v$ vertices, giving (3.2). Between a fixed pair of active sets, Z has $2b^2m$ edges: each edge of H gives two ordered pairs of labels, with b^2 pairs of active vertices for each, where $m = e(H)$. Since $2b^2m \leq b^2vd$ and $a = bv$, at least $a^2(1 - d/v)$ edges remain between each pair of active sets, proving (3.3). $\square$

4 Proof of the main theorem

Proof of Theorem 1. First let $k = 2t + 2$ be even, with $t \geq 3$ fixed. By Alon, Erdős, Holzman, and Krivelevich [1, Theorem 7], for every sufficiently large v there is a K_t -saturated graph H_v on v vertices with maximum degree $d_v = O_t(\sqrt{v})$. Choose odd integers $h_v \sim v^{1/4}$. Then $h_v \rightarrow \infty$ and $h_v d_v / v = O_t(v^{-1/4}) \rightarrow 0$, so all hypotheses of Proposition 4 hold for sufficiently large v .

Apply the proposition and write $h = h_v$, $d = d_v$, $a = 2h_v$, and $N = |V(G_v)|$. The order formula gives $N \sim ta$. Since $e(S) = O_t(h)$, $e(T) = O_t(v)$, and each module has $2a$ edges incident with its active set, the module edges number $O_t(a)$. Hence

$$\binom{t}{2} a^2 \left(1 - \frac{d}{v}\right) \leq e(G_v) \leq \binom{t}{2} a^2 + O_t(a).$$

As $d/v \rightarrow 0$, it follows that

$$\frac{e(G_v)}{N^2} \rightarrow \frac{\binom{t}{2}}{t^2} = \frac{t-1}{2t} = \frac{k-4}{2(k-2)}.$$

For odd $k = 2t + 3 \geq 9$, join each graph in the even construction to K_1 . The resulting graph is k -critical, has $N + 1$ vertices and $e(G_v) + N$ edges, and has the same limiting density, now equal to $(k-5)/(2(k-3))$. $\square$

Code availability

The Lean 4 formalisation is available at <https://github.com/FireflySentinel/erdos-917>.

Declaration of generative AI and AI-assisted technologies in the writing process

GPT-6 Astra proposed the construction of critical graphs and the density estimates, drafted the manuscript, and generated the Lean formalization. GPT-5.6 Sol and Claude Opus 5 were used to organize earlier research material and to review the Lean code. The author checked the arguments, completed the final manuscript, and takes full responsibility for the content.

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